

The Qubit & Superposition

Build a qubit from the ground up: the Bloch sphere, amplitudes, and why a Hadamard gate puts a state in two places at once.

EXAMPLE NOTES

Written by the QuantaVerse team to show what a module's notes can look like: each lesson's theory in brief, every formula typeset and numbered, and the key ones gathered on a formula sheet at the end. Professors can upload their own notes for this module on its page, and students will see them alongside these.

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CONVENTIONS

Kets are column vectors, with $|0\rangle = (1,0)^T$ and $|1\rangle = (0,1)^T$. A dagger is the conjugate transpose, and z^* is the complex conjugate of z . The symbol $\equiv$ between two states means they differ only by a global phase, so no measurement can tell them apart.

LESSON 1.1

A qubit is a direction, not a digit

KEY IDEA

A qubit's state is a unit vector in a two-dimensional complex vector space. The basis vectors of that space are called $|0\rangle$ and $|1\rangle$.

A classical bit holds one of two values. A qubit holds a *direction*: a vector with two complex components. Writing the basis as column vectors makes every later calculation a matter of matrix multiplication.

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (1.1.1)$$

the computational basis

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, \quad \alpha, \beta \in \mathbb{C} \quad (1.1.2)$$

the general state of one qubit

The numbers α and β are the *amplitudes*. They are not free: the vector must have length one, because the squared magnitudes of the amplitudes are the probabilities of the two measurement outcomes, and probabilities sum to one.

$$\langle\psi|\psi\rangle = |\alpha|^2 + |\beta|^2 = 1 \quad (1.1.3)$$

normalisation

DEFINITION · INNER PRODUCT

For $|\phi\rangle = (\phi_0, \phi_1)^\top$ and $|\psi\rangle = (\psi_0, \psi_1)^\top$, the inner product is $\langle\phi|\psi\rangle = \phi_0^*\psi_0 + \phi_1^*\psi_1$. The bra $\langle\phi|$ is the conjugate transpose of $|\phi\rangle$, a row vector. Two states are *orthogonal* when their inner product is zero, as $|0\rangle$ and $|1\rangle$ are.

$$\langle\phi|\psi\rangle = \phi_0^*\psi_0 + \phi_1^*\psi_1, \quad \langle\phi|\psi\rangle^* = \langle\psi|\phi\rangle \quad (1.1.4)$$

the inner product, and its symmetry

Counting parameters shows how much a qubit really holds. Two complex numbers make four real ones. Normalisation removes one, and the global phase (lesson 1.6) removes another, leaving two real parameters: exactly the two angles that place a point on a sphere.

WORKED EXAMPLE · IS IT A VALID STATE?

Take $|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle$. Then $|\alpha|^2 + |\beta|^2 = \frac{1}{2} + |i|^2 \cdot \frac{1}{2} = \frac{1}{2} + \frac{1}{2} = 1$, so it is normalised. Both outcomes have probability one half, and the i is not decoration: it makes this state different from $(|0\rangle + |1\rangle)/\sqrt{2}$.

LESSON 1.2

Amplitudes are not probabilities

KEY IDEA

A probability is the squared magnitude of an amplitude. Amplitudes can be negative or complex, so two routes to the same outcome can cancel, which probabilities never do.

$$P(x) = |\alpha_x|^2 \quad (1.2.1)$$

the probability of reading x is the squared magnitude of its amplitude

Two states can give identical measurement statistics and still be different states. The pair below is the standard example: each gives 0 or 1 with probability one half, yet they are orthogonal, so a suitable measurement tells them apart every time.

$$|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}} \quad (1.2.2)$$

the X-basis states

$$\langle + | - \rangle = \frac{1}{2} (1 \cdot 1 + 1 \cdot (-1)) = 0 \quad (1.2.3)$$

same statistics in the Z basis, yet orthogonal

When an outcome can be reached along two paths, quantum mechanics adds the *amplitudes* first and squares afterwards. Squaring a sum produces a cross term, and that cross term, interference, is the only thing separating a quantum computer from a random-number generator.

$$|a_1 + a_2|^2 = |a_1|^2 + |a_2|^2 + 2 \operatorname{Re}(a_1^* a_2) \quad (1.2.4)$$

interference: the cross term can be positive, negative or zero

PATHS	AMPLITUDES	PROBABILITY OF THE OUTCOME
in step	$a_1 = a_2 = \frac{1}{2}$	$ 1 ^2 = 1$
out of step	$a_1 = \frac{1}{2}, a_2 = -\frac{1}{2}$	$ 0 ^2 = 0$
classical mixture	probabilities $\frac{1}{4} + \frac{1}{4}$	$\frac{1}{2}$, whatever the signs

Every algorithm in this course is a way of arranging these signs so that the paths to wrong answers cancel and the paths to the right answer reinforce.

WORKED EXAMPLE · DESTRUCTIVE INTERFERENCE IN ONE LINE

Two paths reach outcome 1 with amplitudes $+\frac{1}{2}$ and $-\frac{1}{2}$. Adding first gives 0, so $P(1) = 0$. Adding the probabilities instead would have given $\frac{1}{4} + \frac{1}{4} = \frac{1}{2}$. The difference between those two answers is what lesson 1.5 measures.

LESSON 1.3

The Bloch sphere

KEY IDEA

Once normalisation and the global phase are removed, every single-qubit state is described by two angles, which place it on the surface of a unit sphere.

$$|\psi\rangle = \cos\frac{\theta}{2}|0\rangle + e^{i\varphi}\sin\frac{\theta}{2}|1\rangle, \quad 0 \leq \theta \leq \pi, \quad 0 \leq \varphi < 2\pi \quad (1.3.1)$$

the Bloch parametrisation

$$\vec{r} = (\sin\theta \cos\varphi, \sin\theta \sin\varphi, \cos\theta) \quad (1.3.2)$$

the Bloch vector: the point on the sphere

The half-angles look odd at first, but they are what makes the picture work. Orthogonal states, which are 90° apart as vectors, land on *opposite* points of the sphere: $|0\rangle$ at the north pole and $|1\rangle$ at the south.

STATE	θ	φ	BLOCH VECTOR
$ 0\rangle$	0	any	(0,0,1), north pole
$ 1\rangle$	π	any	(0,0,-1), south pole
$ +\rangle$	$\pi/2$	0	(1,0,0)
$ -\rangle$	$\pi/2$	π	(-1,0,0)
$ +i\rangle = (0\rangle + i 1\rangle)/\sqrt{2}$	$\pi/2$	$\pi/2$	(0,1,0)
$ -i\rangle = (0\rangle - i 1\rangle)/\sqrt{2}$	$\pi/2$	$3\pi/2$	(0,-1,0)

The components of the Bloch vector are measurable. Each is the average value of one Pauli observable (lesson 3.1), so the sphere is not just a drawing: three kinds of measurement, repeated, pin the point down.

$$r_x = \langle X \rangle, \quad r_y = \langle Y \rangle, \quad r_z = \langle Z \rangle \quad (1.3.3)$$

each coordinate is an expectation value

$$P(0) = \cos^2\frac{\theta}{2} = \frac{1+r_z}{2}, \quad P(1) = \sin^2\frac{\theta}{2} = \frac{1-r_z}{2} \quad (1.3.4)$$

measurement probabilities read off the height

The azimuth φ never affects $P(0)$ or $P(1)$. It is a relative phase, invisible to a Z-basis measurement, but it decides what a Hadamard or any other rotation does next.

WORKED EXAMPLE · WHERE IS IT?

For $|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}e^{i\pi/4}|1\rangle$: $\cos\frac{\theta}{2} = \frac{\sqrt{3}}{2}$ gives $\theta = \pi/3$, and $\varphi = \pi/4$. So

$$\vec{r} = \left(\frac{\sqrt{3}}{2} \cdot \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2} \cdot \frac{\sqrt{2}}{2}, \frac{1}{2}\right) \approx (0.61, 0.61, 0.5) \text{ and } P(0) = \frac{1+0.5}{2} = \frac{3}{4}.$$

LESSON 1.4

The Hadamard gate

KEY IDEA

H exchanges the Z basis and the X basis. Applied twice it does nothing, which is what lets it create a superposition and then undo it.

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad (1.4.1)$$

the Hadamard gate

$$H|0\rangle = |+\rangle, \quad H|1\rangle = |-\rangle, \quad H|+\rangle = |0\rangle, \quad H|-\rangle = |1\rangle \quad (1.4.2)$$

its action on the two bases

$$H^2 = I, \quad H^\dagger = H \quad (1.4.3)$$

self-inverse and Hermitian

On the Bloch sphere H is a half-turn about the axis halfway between x and z . The half-turn swaps the z axis with the x axis, which is the same statement as the basis swap above.

A compact form covers both inputs at once and generalises to many qubits, where it becomes the workhorse of every algorithm in this course.

$$H|x\rangle = \frac{1}{\sqrt{2}} \sum_{z \in \{0,1\}} (-1)^{xz} |z\rangle, \quad x \in \{0,1\} \quad (1.4.4)$$

one formula for both basis states

$$H^{\otimes n} |0\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle \quad (1.4.5)$$

n Hadamards: an equal superposition of all 2^n strings

WORKED EXAMPLE · MATRIX CHECK

$$H|1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = |-\rangle.$$

$$\text{And } H^2 = \frac{1}{2} \begin{pmatrix} 1+1 & 1-1 \\ 1-1 & 1+1 \end{pmatrix} = I, \text{ so a second } H \text{ always returns the qubit to where it started.}$$

LESSON 1.5

Superposition is not “both at once”

KEY IDEA

A superposition is one definite state, not a hidden coin toss. Applying H twice gives a result no random process can reproduce.

A coin model of H would say: send in 0, get out 0 or 1 at random. That model predicts a single H correctly, since $P(0) = P(1) = \frac{1}{2}$. Now apply H twice. The coin model says the second toss is random too, so the result stays 50/50. The quantum result is 0 every time.

$$HH|0\rangle = H \frac{|0\rangle + |1\rangle}{\sqrt{2}} = \frac{1}{2} \left[(|0\rangle + |1\rangle) + (|0\rangle - |1\rangle) \right] = |0\rangle \quad (1.5.1)$$

the two-Hadamard experiment

PATH TO THE FINAL OUTCOME	AMPLITUDE VIA $ 0\rangle$	AMPLITUDE VIA $ 1\rangle$	TOTAL
ends at 0	$\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2}$	$\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2}$	1
ends at 1	$\frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2}$	$\frac{1}{\sqrt{2}} \cdot \left(-\frac{1}{\sqrt{2}}\right) = -\frac{1}{2}$	0

The difference between a superposition and a coin can be written down precisely using density matrices. A coin, a *mixture* of 0 and 1, has no off-diagonal entries. The state $|+\rangle$ does. Those off-diagonal entries, the *coherences*, are what interference needs.

$$\rho_{\text{coin}} = \frac{1}{2} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \rho_+ = |+\rangle\langle+| = \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \quad (1.5.2)$$

same diagonal, different off-diagonals

$$H\rho_{\text{coin}}H^\dagger = \rho_{\text{coin}}, \quad H\rho_+H^\dagger = |0\rangle\langle 0| \quad (1.5.3)$$

a Hadamard cannot sharpen a coin, but it can sharpen a superposition

Decoherence is the loss of those off-diagonal terms through contact with the environment. It is why a real device slowly turns ρ_+ into ρ_{coin} , and why circuits have to finish quickly (lesson 5.7).

LESSON 1.6

Global phase, and what it is not

KEY IDEA

Multiplying a whole state by $e^{i\gamma}$ changes nothing that can be measured. A phase between components changes the state.

$$e^{i\gamma}|\psi\rangle \equiv |\psi\rangle$$

(1.6.1)

global phase: physically the same state

The reason is one line: every probability is a squared magnitude, and $|e^{i\gamma}| = 1$. The same holds for measurements in any basis, so no experiment separates the two.

$$|\langle b|e^{i\gamma}\psi\rangle|^2 = |e^{i\gamma}|^2 |\langle b|\psi\rangle|^2 = |\langle b|\psi\rangle|^2 \quad (1.6.2)$$

for every basis state $|b\rangle$ of every measurement

$$\alpha|0\rangle + e^{i\varphi}\beta|1\rangle$$

(1.6.3)

relative phase φ : a rotation about z on the Bloch sphere, and a different state

A relative phase is invisible in the Z basis but not in others. The states $|+\rangle$ and $|-\rangle$ differ only by the relative phase $\varphi = \pi$, and a Hadamard sends them to $|0\rangle$ and $|1\rangle$, which a measurement separates perfectly.

$$-|\psi\rangle \equiv |\psi\rangle, \quad \text{but} \quad \frac{|0\rangle - |1\rangle}{\sqrt{2}} \not\equiv \frac{|0\rangle + |1\rangle}{\sqrt{2}} \quad (1.6.4)$$

a global sign versus a relative sign

The same freedom applies to gates: U and $e^{i\gamma}U$ are the same operation. That is why gate identities are often only true up to a phase.

$$XZ = -iY, \quad ZX = iY$$

(1.6.5)

 XZ and ZX are the same gate, up to a global phase

One caution for later: A global phase stops being global when the gate is *controlled*: controlled- $(e^{i\gamma}U)$ applies the phase only when the control is 1, turning it into a relative phase on the control qubit. That is exactly the phase kickback of lesson 6.3.

WORKED EXAMPLE · GLOBAL OR RELATIVE?

$\frac{1}{\sqrt{2}}(i|0\rangle - i|1\rangle) = i \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle) = i|-\rangle$: the factor i is global, so the state is $|-\rangle$.

$\frac{1}{\sqrt{2}}(|0\rangle - i|1\rangle)$ has no common factor to pull out, so its $-i = e^{i3\pi/2}$ is a relative phase, and the state is $|-i\rangle$, not $|-\rangle$.

MODULE 1 · FORMULA SHEET

Formula sheet

The key results of the module, numbered as they appear in the lessons.

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n Hadamards: an equal superposition of all 2^n strings

$$H^{\otimes n} |0\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{x=0}^{2^n-1} |x\rangle \quad (1.4.5)$$

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